Informatica® Metadata Command Center
October 2026

Microsoft Azure Synapse Data Warehouse Sources

Informatica Metadata Command Center Microsoft Azure Synapse Data Warehouse Sources
October 2026

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Table of Contents

Preface 5

Chapter 1: Introduction to Microsoft Azure Synapse Data Warehouse catalog sources 6

Extraction and view process 7

About the Microsoft Azure Synapse Data Warehouse catalog source 8

- Extracted metadata 8
- Data profiling for Microsoft Azure Synapse Data Warehouse objects 9
- Compatible connectors 9

Chapter 2: Before you begin 10

Verify permissions 10

- Permissions to extract metadata 10
- Permissions to run data profiles 10
- Permissions to perform data classification 11
- Permissions to perform relationship discovery 11
- Permissions to perform glossary association 11

Create a connection 11

- Storage types 14
- Azure Blob storage 14
- ADLS Gen2 storage 15

Create endpoint catalog sources for connection assignment 17

Import a relationship inference model 17

Chapter 3: Create catalog sources in Metadata Command Center 18

Step 1. Register a catalog source 18

Step 2. Configure capabilities 20

- Configure metadata extraction 20
- Configure data profiling and quality 22
- Configure data observability 26
- Configure data classification 28
- Configure relationship discovery 29
- Configure glossary associations 29
- Configure automation 30

Step 3. Associate stakeholders and asset groups 32

Step 4. Run or schedule the job 34

Step 5. Assign reference catalog source connections to endpoint catalog source objects 35

Chapter 4: View results in Data Governance and Catalog 37

View metadata extraction results 37

View data lineage. 39
View lineage at the catalog source level. 39
View lineage at the data set level. 40
View lineage at the data element level. 40
View data profiling results 40
View data observability results 41
View classified data. 42
View glossary associations. 42

Appendix A: Managing catalog sources. 44

Cloning a catalog source configuration. 44
Deleting or purging a catalog source. 45
Purging obsolete objects from a catalog source. 45
Curating CLAIRE recommendations. 46
Managing configuration permissions for catalog sources. 46

Preface

Read *Microsoft Azure Synapse Data Warehouse Sources* to learn how to register and configure Microsoft Azure Synapse Data Warehouse sources as catalog sources in Metadata Command Center. After you configure a catalog source, you extract metadata and then view the results in Data Governance and Catalog.

CHAPTER 1

Introduction to Microsoft Azure Synapse Data Warehouse catalog sources

You can use Metadata Command Center to extract metadata from a source system.

A source system is any system that contains data or metadata. For example, Microsoft Azure Synapse Data Warehouse is a source system from which you can extract metadata through a Microsoft Azure Synapse Data Warehouse catalog source. A catalog source is an object that represents and contains metadata from the source system.

Before you extract metadata from a source system, you first create and register a catalog source that represents the source system. Then you configure capabilities for the catalog source. A capability is a task that Metadata Command Center can perform, such as metadata extraction, lineage discovery, autonomous enrichment, data profiling, data classification, or glossary association.

When Metadata Command Center extracts metadata, Data Governance and Catalog displays the extracted metadata and its attributes as technical assets. You can then perform tasks such as analyzing the assets, viewing lineage, and creating links between those assets and their business context.

The following table describes the capabilities of the catalog source:

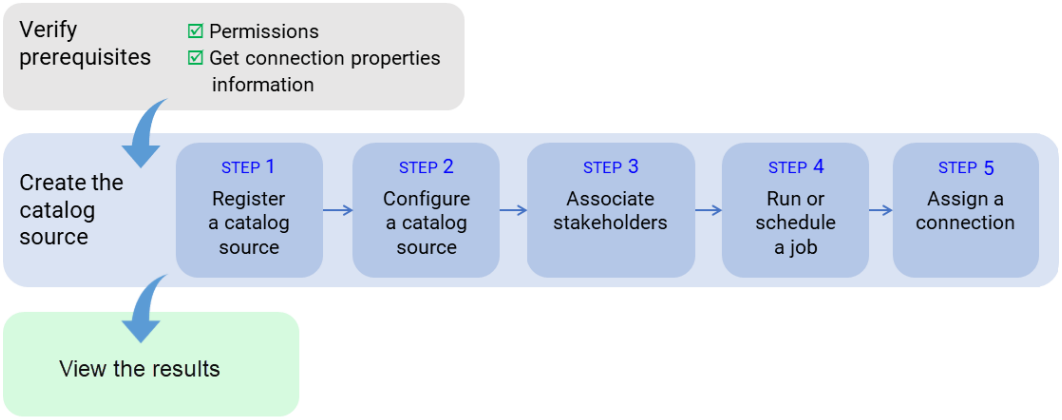
Capability	Description
Advanced Programming Language Parsing	Advanced Programming Language Parsing parses the source system code in addition to extracting objects from the source system.
Incremental metadata extraction	An incremental metadata extraction extracts only the changed and new objects since the last catalog source job run. Incremental metadata extraction doesn't remove deleted objects from the catalog and doesn't extract metadata of code-based objects if applicable.
Serverless Runtime Environment	A serverless runtime environment is an advanced serverless deployment solution that doesn't require downloading, installing, configuring, or maintaining a Secure Agent or Secure Agent group. You can use a serverless runtime environment in the same way that you use a Secure Agent when you configure a catalog source.
Lineage Discovery	Builds the complete lineage of a catalog source by recommending endpoint catalog source objects to assign to reference catalog source connections. When you run the catalog source job, Metadata Command Center assigns the reference catalog source connections to CLAIRE recommended endpoint catalog source objects. You can then view the list of CLAIRE recommendations and accept or reject them.

Capability	Description
Data Profiling and Quality	- **Data Profiling**. Assesses source metadata and analyzes the collected statistics to discover content and structure, such as value distribution, patterns, and data types. - **Data Quality**. Measures the reliability of the data and enables data usage.
Data Observability	Identifies anomalies in the characteristics of the data that you profile.
Data Classification	Data classification is the process of identifying and organizing data into relevant categories based on the functional meaning of the data. Classifying data can help your organization manage risks, compliance, and data security.
Relationship Discovery	You can associate terms that are in the glossary with technical assets to provide user-friendly business names to technical assets. Glossary Association automatically associates glossary terms with technical assets or recommends glossary terms that you can manually associate with technical assets in Data Governance and Catalog.
Glossary Association	You can associate terms that are in the glossary with technical assets to provide user-friendly business names to technical assets. Glossary Association automatically associates glossary terms with technical assets or recommends glossary terms that you can manually associate with technical assets in Data Governance and Catalog.
Autonomous Enrichment	Automatically propagates classification, description, business name, and glossary enrichments from an asset to related assets based on the lineage between assets. Autonomous enrichment reduces manual curation effort by extending the enrichments that you apply to one asset across assets that are connected to it through lineage.

Extraction and view process

To extract metadata from a source system, configure the catalog source and run the extraction job in Metadata Command Center. Then view the results in Data Governance and Catalog.

The following image shows the process to extract metadata from a source system:



After you verify prerequisites, perform the following tasks to extract metadata from Microsoft Azure Synapse Data Warehouse:

1. Register a catalog source. Create a catalog source object, select Microsoft Azure Synapse Data Warehouse, and then select and test the connection.
2. Configure the catalog source. Specify the runtime environment and configure parameters for metadata extraction. Optionally, add filters to include or exclude source system assets from metadata extraction. You can also configure other capabilities such as lineage discovery, autonomous enrichment, data profiling and quality, data classification, or glossary association.
3. Optionally, associate stakeholders. Associate users with technical assets, giving the users permission to perform actions determined by their roles.
4. Run or schedule the catalog source job.
5. Optionally, if the catalog source job generates referenced asset objects, you can assign a connection to referenced source system assets.
   You can view the lineage with object references without performing connection assignment. After connection assignment, you can view the objects.

After you run the catalog source job, you view the results in Data Governance and Catalog.

About the Microsoft Azure Synapse Data Warehouse catalog source

You can use the Microsoft Azure Synapse Data Warehouse catalog source to extract metadata from a Microsoft Azure Synapse Data Warehouse source system.

Microsoft Azure Synapse Data Warehouse, formerly known as Azure SQL Data Warehouse, is a big data analytic service that queries and analyzes your data. Create a Microsoft Azure Synapse Data Warehouse catalog source to access data from Microsoft Azure SQL Data Warehouse.

Extracted metadata

You can extract specific metadata from a Microsoft Azure Synapse Data Warehouse source system with a Microsoft Azure Synapse Data Warehouse catalog source.

Metadata Command Center extracts the following metadata from a Microsoft Azure Synapse Data Warehouse source system:

- Database
- Table
- Schema
- View
- Materialized View
- Function
- Stored Procedure
- Column
- External Table
- External Column

Data profiling for Microsoft Azure Synapse Data Warehouse objects

Configure data profiling to run profiles on the metadata extracted from a Microsoft Azure Synapse Data Warehouse source system. You can view the profiling statistics in Data Governance and Catalog.

You can run data profiles on the following objects:

- Tables
- External Tables created in the following file formats on the Microsoft Azure Serverless SQL pool:
  - Delimited text
  - Parquet
- Views

The data profiling task runs profiles on the following data types:

- Bigint
- Bit
- Char
- Date
- Datetime
- Datetime2
- Decimal
- Float
- Int
- Nchar
- Nvarchar
- Real
- Smalldatetime
- Smallint
- Time
- Tinyint
- Uniqueidentifier
- Varchar

Compatible connectors

Before you configure a Microsoft Azure Synapse Data Warehouse catalog source, you must connect to the Microsoft Azure Synapse Data Warehouse source system.

Use the Microsoft Azure Synapse SQL connector to connect to the Microsoft Azure Synapse Data Warehouse source system.

For information about configuring a connection, see *Connections* in the Administrator help.

CHAPTER 2

Before you begin

Before you create a catalog source, ensure that you have the information required to connect to the source system.

Perform the following tasks:

- Assign the required permissions.
- Configure a connection to the Microsoft Azure Synapse Data Warehouse source system in Administrator.
- Create endpoint catalog sources for connection assignment.
- Optionally, if you want to identify pairs of similar columns and relationships between tables within a catalog source, import a relationship inference model.

Verify permissions

To extract metadata and to configure other capabilities that a catalog source might include, you need account access and permissions on the source system. The permissions required might vary depending on the capability.

Permissions to extract metadata

Ensure that you have the required permissions to enable metadata extraction.

Configure the `VIEW DEFINITION` and `CONNECT` permission for the user account to view all schemas from where you can load metadata.

For information about configuring permissions for the user account, see the Microsoft Azure Synapse documentation.

Permissions to run data profiles

Ensure that you have the required permissions to run profiles.

Grant the `db_owner` privilege or the following more granular privileges to the user to connect to Microsoft Azure Synapse Data Warehouse:

- `EXEC sp_addrolemember 'db_datareader', '<user>';` // Alternately assign permission to the individual table.
- `EXEC sp_addrolemember 'db_datawriter', '<user>';` // Alternately assign permission to the individual table.

- `GRANT ALTER ANY EXTERNAL DATA SOURCE TO <user>;`
- `GRANT ALTER ANY EXTERNAL FILE FORMAT TO <user>;`
- `GRANT CONTROL TO <user>;` // To grant all permissions on the database.

  Note: You need this permission to create and access the external data source. The Microsoft Azure Synapse SQL connector uses the Polybase command to load data from external data sources. Polybase requires external tables to reference external data.

  or

  `GRANT ALTER ON SCHEMA::<schema_name> TO <user>;` // To grant permissions only on the specified schema.

  Note: Verify that you configured the External Data Source property in the advanced connection properties for Microsoft Azure Synapse SQL Connector.
- `GRANT CREATE TABLE TO <user>;`

Permissions to perform data classification

You can perform data classification with the permissions required on the source system to perform metadata extraction.

Permissions to perform relationship discovery

You can perform relationship discovery with the permissions required on the source system to perform metadata extraction.

Permissions to perform glossary association

You can perform glossary association with the permissions required on the source system to perform metadata extraction.

Create a connection

Before you configure the Microsoft Azure Synapse Data Warehouse catalog source, create a connection object in Administrator.

1. In Administrator, select **Connections**.
2. Click **Add Connection**.
3. Select Microsoft Azure Synapse SQL and click **Next**.
4. Enter the connection properties.

The following table describes the connection properties:

Property	Description
Connection Name	Name of the connection. Each connection name must be unique within the organization. Connection names can contain alphanumeric characters, spaces, and the following special characters: _ . + -, Maximum length is 255 characters.
Description	Description of the connection. Maximum length is 4000 characters.
Use Secret Vault	Stores sensitive credentials for this connection in the secrets manager that is configured for your organization. This property appears only if secrets manager is set up for your organization. When you enable the secret vault in the connection, you can select the credentials that the Secure Agent retrieves from the secrets manager. If you don't enable this option, the credentials are stored in the repository or on a local Secure Agent, depending on how your organization is configured. **Note:** If you're using this connection to apply data access policies through pushdown or proxy services, you cannot use the Secret Vault configuration option. For information about how to configure and use a secrets manager, see "Secrets manager configuration" in the Administrator help.
Runtime Environment	The name of the runtime environment where you want to run tasks. Select a Secure Agent, Hosted Agent, or serverless runtime environment. Hosted Agent doesn't apply to mappings in advanced mode. For more information about runtime environments, see *Runtime Environments* in the Administrator help.

Property	Description

Azure DW JDBC URL	The Microsoft Azure Synapse SQL JDBC connection string.
	Use the following string to connect to Microsoft Azure Synapse SQL:
	`jdbc:sqlserver://<Server>.database.windows.net:1433;database=<Database>`
	You can include an authentication parameter in the connection string to specify the authentication type. You can configure the following authentication types to connect to Microsoft Azure Synapse SQL:
	- Microsoft SQL Server
	- Azure Active Directory
	- Managed Identity
	- Service Principal
	If you don't include an authentication parameter in the connection string, the Secure Agent uses Microsoft SQL Server authentication as the authentication type.
	Use the following string to connect to a serverless SQL pool in Microsoft Azure Synapse SQL:
	`jdbc:sqlserver://<Serverless SQL endpoint>:1433;database=<Database>;Authentication=ActiveDirectoryMsi;`
	Connection string format for Microsoft SQL Server authentication
	`jdbc:sqlserver://<Server>.database.windows.net:1433;database=<Database>`
	Connection string format for Azure Active Directory (AAD) authentication
	`jdbc:sqlserver://<Server>.database.windows.net:1433;database=<Database>;encrypt=true;trustServerCertificate=false;hostNameInCertificate=*.database.windows.net;loginTimeout=30;Authentication=ActiveDirectoryPassword;`
	Connection string format for Service Principal authentication
	`jdbc:sqlserver://<Server>.database.windows.net:1433;database=<Database>;encrypt=true;trustServerCertificate=false;hostNameInCertificate=*.database.windows.net;loginTimeout=30; Authentication=ActiveDirectoryServicePrincipal;`
	Connection string format for Managed Identity authentication
	`jdbc:sqlserver://<Server>.database.windows.net:1433;database=<Database>;Authentication=ActiveDirectoryMsi;`

Property	Description
Azure DW JDBC Username	User name to connect to the Microsoft Azure Synapse SQL account. - For AAD authentication, provide your AAD user name. - For Microsoft SQL server authentication, provide your SQL auth user name. - For service principal authentication, provide the application ID or client ID for your application registered in Azure Active Directory. This property doesn't apply to Managed Identity authentication.
Azure DW JDBC Password	Password to connect to the Microsoft Azure Synapse SQL account. - For AAD authentication, provide the password of the AAD user. - For Microsoft SQL server authentication, provide the password of SQL auth user. - For service principal authentication, provide the client secret for your application registered in the Azure Active Directory. This property doesn't apply to Managed Identity authentication.
Azure DW Client ID	Required if you want to use the user-assigned managed identity for Managed Identity Authentication to connect to Microsoft Azure Synapse SQL. The client ID of the user-assigned managed identity. If you use system-assigned managed identity, leave the field empty.
Azure DW Schema Name	Name of the schema in Microsoft Azure Synapse SQL.

5. Select the storage type to stage the files.
6. Click **Test** to test your connection to the source system.
7. Click **Save**.

Storage types

You can select Microsoft Azure Blob Storage or Microsoft Azure Data Lake Storage Gen2 as the Azure storage type to stage the data files. Default is Azure Blob.

Select your preferred storage type and then configure the storage-specific parameters.

To get credentials for shared key authentication when you connect to Microsoft Azure Blob Storage or Microsoft Azure Data Lake Storage Gen2 to stage files, see Get credentials for shared key authentication How-To Library article.

To get credentials for service principal authentication when you connect to Microsoft Azure Data Lake Storage Gen2 to stage files, see Get credentials for service principal authentication How-To Library article.

Azure Blob storage

When you select Microsoft Azure Blob as the storage type, you can configure Shared Key Authentication as the authentication type to stage the files.

Note: If you connect to a serverless SQL pool, you must configure Microsoft Azure Data Lake Storage Gen2 as the storage type.

The following table describes the authentication type that you can configure for Microsoft Azure Blob storage:

Property	Description
Authentication Type	Authentication type to connect to Microsoft Azure Blob storage to stage the files. You can configure Shared Key Authentication as the authentication type to stage the files.

Shared Key Authentication

Uses the storage account name and account key to connect to Microsoft Azure Blob storage.

The following table describes the basic connection properties for shared key authentication:

Property	Description
Azure Blob Account Name	Name of the Microsoft Azure Blob Storage account to stage the files.
Azure Blob Account Key	The Microsoft Azure Blob Storage access key to stage the files.
Container Name	The name of the container in the Azure Blob Storage account.

ADLS Gen2 storage

When you select Microsoft Azure Data Lake Storage Gen2 as the storage type, you can configure various authentication types to stage the files.

The following table describes authentication types that you can configure for Microsoft Azure Data Lake Storage Gen2 storage:

Property	Description
Authentication Type	Authentication type to connect to Azure storage to stage the files. Select one of the following options: - Shared Key Authentication - Service Principal Authentication - Managed Identity Authentication For more information on how to configure the authentication types, see Setting up authentication to connect to Microsoft Azure Synapse SQL.

Shared Key Authentication

Uses the storage account name and account key to connect to Microsoft Azure Data Lake Storage Gen2.

Note: You cannot select shared key authentication type when you connect to a serverless SQL pool.

The following table describes the basic connection properties for shared key authentication:

Property	Description
ADLS Gen2 Storage Account Name	Name of the Microsoft Azure Data Lake Storage Gen2 account to stage the files.
ADLS Gen2 Account Key	The Microsoft Azure Data Lake Storage Gen2 access key to stage the files.
File System Name	The name of the file system in the Microsoft Azure Data Lake Storage Gen2 account.

Service Principal Authentication

Uses the account name, client ID, client secret, and tenant ID to connect to Microsoft Azure Data Lake Storage Gen2.
The following table describes the basic connection properties for service principal authentication:

Property	Description
ADLS Gen2 Storage Account Name	Name of the Microsoft Azure Data Lake Storage Gen2 account to stage the files.
Client ID	The client ID of your application. Enter the application ID or client ID for your application registered in the Azure Active Directory.
Client Secret	The client secret for your application.
Tenant ID	The directory ID or tenant ID for your application.
File System Name	The name of the file system in the Microsoft Azure Data Lake Storage Gen2 account.

Managed Identity Authentication

Select this authentication type to authenticate using system-assigned or user-assigned identities that are assigned to applications in Azure to access Azure resources in Microsoft Azure Data Lake Storage Gen2.

The following table describes the basic connection properties for managed identity authentication:

Property	Description
ADLS Gen2 Storage Account Name	Name of the Microsoft Azure Data Lake Storage Gen2 account to stage the files.
Client ID	The client ID of your application. Enter the client ID for the user-assigned managed identity. If the managed identity is system-assigned, leave the field empty.
File System Name	The name of the file system in the Microsoft Azure Data Lake Storage Gen2 account.

Create endpoint catalog sources for connection assignment

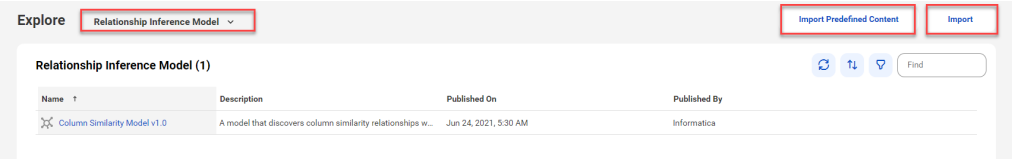
An endpoint catalog source represents a source system that the catalog source references. Before you perform connection assignment, create endpoint catalog sources and run the catalog source jobs.

You can then perform connection assignment to reference source systems to view complete lineage with source system objects.

Import a relationship inference model

Import a relationship inference model if you want to configure the relationship discovery capability. You can either import a predefined relationship inference model, or import a model file from your local machine.

1. In Metadata Command Center, click **Explore** on the navigation panel.
2. Expand the menu and select **Relationship Inference Model**. The **Explore** page opens with the **Relationship Inference Model** menu.



3. Select one of the following options:
   - **Import Predefined Content**. Imports a predefined relationship inference model called Column Similarity Model v1.0.
   - **Import**. Imports the predefined relationship inference model from your local machine. Select this if you previously imported predefined content into your local machine and the inference model is stored on the machine.
     To import a file, click **Choose File** in the **Import Relationship Inference Model** window and navigate to the model file on your local machine. You can also drag and drop the file.

   The imported models appear in the list of relationship inference models on the **Relationship Discovery** tab.

CHAPTER 3

Create catalog sources in Metadata Command Center

Use Metadata Command Center to configure a catalog source for Microsoft Azure Synapse Data Warehouse and run the catalog source job.

When you configure a catalog source, you define the source system that you want to extract metadata from. Configure filters to include or exclude source system metadata before you run the job. Optionally, configure other capabilities, such as lineage discovery, autonomous enrichment, data profiling and quality, data classification, relationship discovery, and glossary association.

To provide stakeholders access to technical assets, you can assign access through stakeholder roles. You can also associate technical assets extracted from the catalog source to asset groups. If your catalog source references other source systems, you can create a connection assignment to the endpoint catalog source to view complete lineage.

Step 1. Register a catalog source

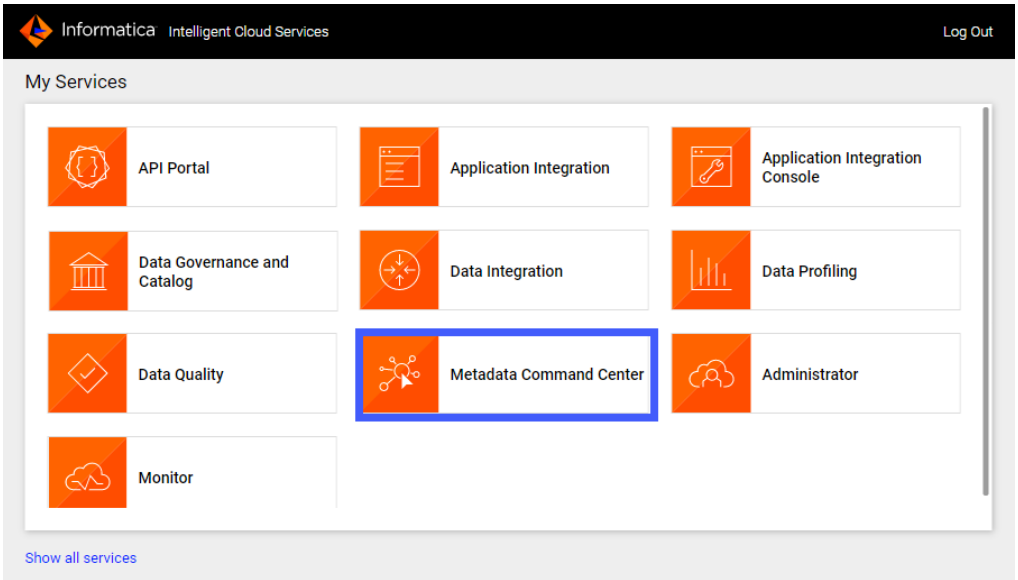
When you register a catalog source, provide general information and connection values.

1. Log in to Intelligent Data Management Cloud.

   The **My Services** page appears.

2. Click **Metadata Command Center**.

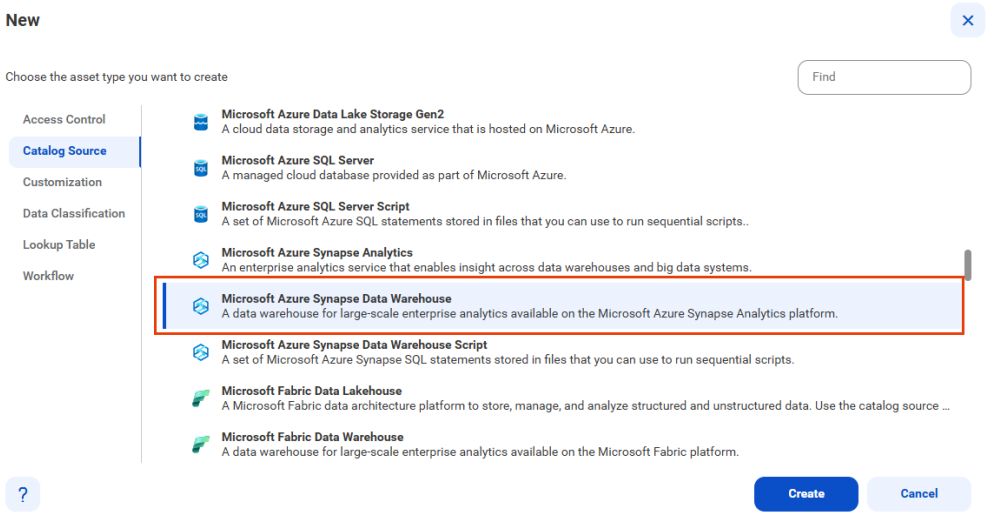
   The **Metadata Command Center** box appears on the **My Services** page.



The Metadata Command Center home page appears.

3. Click **New**.
4. Select **Catalog Source** from the list of asset types.
5. Select Microsoft Azure Synapse Data Warehouse from the list of catalog source types.

   The following image shows the Microsoft Azure Synapse Data Warehouse catalog source:



6. Click **Create**.

   The **New Catalog Source** page opens.
7. In the **General Information** section, enter a name and an optional description for the catalog source.

   Note: You can rename a catalog source after you create it, but to apply the change to all associated objects you must rerun the metadata extraction job.

   After you save the catalog source, you can update the description in Metadata Command Center and Data Governance and Catalog. The update appears only in the service in which you update it.
8. In the **Connection Information** area, select the connection that you created in Administrator.

Note: To create or edit a catalog source, you need permissions on the connection to the source system. Select a connection that you have access to, or ask the administrator to grant the necessary permissions to the connection that you want to use.

9. Click **Connection Properties** to expand and view the connection properties for the selected connection.
10. Click **Test Connection** to test your connection to the source system.
11. Click **Next**.

The **Configuration** page appears.

Step 2. Configure capabilities

When you configure the Microsoft Azure Synapse Data Warehouse catalog source, you define the settings for the metadata extraction capability and other optional capabilities.

The metadata extraction capability extracts source metadata from external source systems. You can also configure other capabilities that the catalog source includes.

Configure metadata extraction

When you configure the Microsoft Azure Synapse Data Warehouse catalog source, you choose a runtime environment, define filters, and enter configuration parameters for metadata extraction.

1. In the **Connection and Runtime** area, choose a serverless runtime environment or the Secure Agent group where you want to run catalog source jobs.

   Note: Serverless runtime environment options are available if the catalog source works with a serverless runtime environment.

2. Choose to retain, delete, or deprecate objects that are deleted from the source system in the catalog with the **Metadata Change Option**.

   - **Retain**. Retains objects that are deleted from the source system in the catalog. If you update or add a filter, the catalog retains objects extracted from the previous job and extracts additional objects that match the current filter. Objects deleted from the source system are not deleted from the catalog. Enrichments added on deleted objects and relationships are retained.
   - **Delete**. Deletes metadata from the catalog based on objects deleted from the source system and changes you make to the filter. Enrichments added on deleted objects and relationships are also permanently lost. Objects renamed in the source system are removed and recreated in the catalog.
   - **Deprecate**. When objects are deleted from the source system, the catalog changes the lifecycle of the imported objects to Obsolete. The lifecycle of objects also changes to Obsolete when you add exclusion filters or reduce the scope so that objects extracted in earlier jobs are no longer included. For example, the first job includes the sales schema. The second job also includes the sales schema, but excludes the sales.transactions table. In this case, the second job marks the sales.transactions object as Obsolete, even though the first job includes it. This does not impact enrichments applied to deprecated objects and relationships. Objects renamed in the source system are removed and recreated in the catalog. When you run the catalog source job again for other capabilities such as data classification, relationship discovery, or glossary association, the job doesn't consider obsolete objects. Obsolete objects remain in the catalog until they are purged when you run a **Purge Obsolete Objects** job on the **Explore** page.

   Note: You can also change the configured metadata change option when you run a catalog source.

3. In the **Filters** area, define one or more filter conditions to apply for metadata extraction:

   To define filters, you can either select an object type and enter the path to the object as the filter value, or select an object from a list of objects available in the source system.

   a. Select **Yes** to view filter options.

   b. From the Include/Exclude list, choose to include or exclude metadata based on the filter parameters.

   c. Perform one of the following steps:

      - From the Object type list, select an object type or select **All**, depending on the object that you want to extract metadata from. Enter the path to the object as the filter value.
      - In the filter value field, click the Search button and select an object from a list of objects available in the source system.
        The Object type field updates based on the selected object.

        If you select an object type and then click the Search button, the list of objects includes all object types, but you can only select objects that match the selected object type.
        You can edit the filter value after you select an object from the list.

        Note: You can only search for object types that work with the search functionality. If you don't see the Search button for the selected object, enter the object path as the filter value.

        Note: If the object metadata is available in Data Governance and Catalog, a check mark appears next to the object.

        Note: To select an object, you need to have permissions on the connection to the source system.

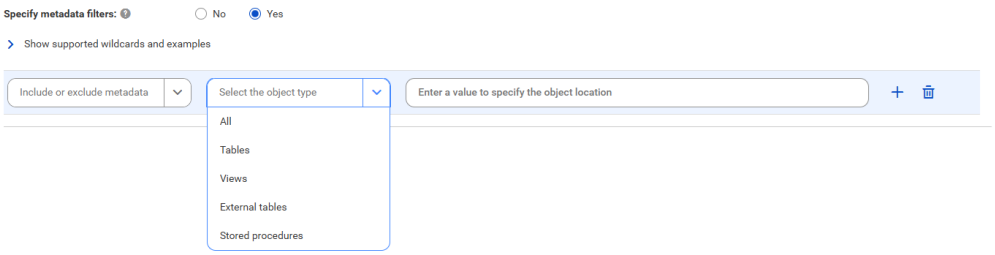
   d. Optionally, to define an additional filter with an OR condition, click the **Add** icon.

   Filters can contain the following wildcards:

   - Question mark. Represents a single character.
   - Asterisk. Represents multiple characters or empty text.

   For object hierarchies, use a dot as a separator. When you enter values for filters, enclose them in double quotes if you use a space or a dot in a single segment.

   The following image shows the filter condition options:



   Examples:

   - To include or exclude all object types in the 'Schema1' schema in the 'dbName1' database, select **All** and enter `dbName1.Schema1` in the value field.
   - To include or exclude all stored procedures in the 'dbName1' database, select **Stored procedures** and enter `dbName1` in the value field.
   - To include or exclude all tables in databases with names that start with 'dbName' followed by one additional character, select **Tables** and enter `dbName?` in the value field.

4. Optional. In the **Configuration Parameters** area, enter properties to override default content values and job parameters. Click **Show Advanced** to view all configuration parameters.

   The following table describes the properties that you enter for Catalog Source Configuration options:

Parameter	Description
Default variables values	Specify a default value for variables used in the programmable objects.
MetaTables Include Filter	Advanced parameter. When you process PL/SQL statements, Metadata Command Center does not read tables or view content by default. If you want to use the content, for example, to process dynamic SQL statements, use the **MetaTables Include Filter** parameter. This parameter prompts the database for the required metadata. Verify that the user has SELECT permissions for metatables. **Note:** Don't use this option to specify filters for tables that you want to include or exclude during the metadata extraction run.

5. Optional. In the **Configuration Parameters** area, enter additional settings.

   The following table describes the property that you enter for additional settings:

   Note: The **Additional Settings** section appears when you click **Show Advanced**.

Property	Description
Expert Parameters	Enter additional configuration options to be passed at runtime. Required if you need to troubleshoot the catalog source job. **Caution:** Use expert parameters when it is recommended by Informatica Global Customer Support.

6. Configure additional capabilities for the catalog source by clicking on the tabs.

Configure data profiling and quality

Enable the data profiling capability to evaluate the quality of metadata extracted from the Microsoft Azure Synapse Data Warehouse source system.

1. Click the **Data Profiling and Quality** tab.
2. Expand **Data Profiling** and select **Enable Data Profiling**.

   Note: Ensure that you have permissions on all the staging connections that you use in your data profiling configuration. You can't run the job if you don't have permissions on the connections that you use. Select connections that you have access to, or ask the administrator to grant the necessary permissions on the connections that you want to use.
3. In the **Connection and Runtime** area, choose the Secure Agent group where you want to run catalog source jobs.
4. Optionally, specify data profiling filters to run the profile on a subset of the metadata that you extract.
   a. Select **Yes** to view filter options.
   b. From the Include/Exclude list, choose to include or exclude metadata based on the filter parameters.

c. From the Object type list, select **Tables**, **Views**, or **External Tables** depending on the object that you want to profile. Select **All** to profile all objects.

d. Enter a value to specify the object location.

Examples:

- You extracted metadata from multiple schemas and now you want to run a profile on all objects in a specific schema. Select **Include Metadata** from the Include/Exclude list, **All** from the Object type list, and then enter the schema name in the input field. For example, `Schema1`
- You extracted metadata of all tables and views from Schema1 schema and now you want to profile all tables in the schema with names that start with Table followed by one additional character. Select **Include Metadata** from the Include/Exclude list, **Tables** from the Object type list, and then enter `Schema1.Table?` in the input field.

To include or exclude multiple objects, click the **Add** icon to add filters with the OR condition.

5. In the **Parameters** area, configure the parameters.

The following table describes the parameters that you can enter:

Parameter	Description
Modes of Run	Determine the type of data that you want the data profiling task to collect. Choose one of the following options: • **Keep Signatures Only**. Collects only aggregate information such as data types, average, standard deviation, and patterns. • **Keep Signatures and Values**. Collects both signatures and data values.
Profiling Scope	Determine whether you want to run data profiling only on the changes made to the source system or on the entire source system. Choose one of the following options: • **Incremental**. Includes only source metadata that is changed or updated since the last profile run. • **Full**. Includes the entire metadata that is extracted based on the filters applied for extraction.
Sampling Type	Determine the sample rows on which you want to run the data profiling task. Choose one of the following options: • **All Rows**. Runs data profiling on all rows in the metadata. • **Limit N Rows**. Runs data profiling on a limited number of rows. • **Random N Rows**. Runs data profiling on the selected number of random rows.
No of rows to limit	Required if you select **Limit N Rows** or **Random N Rows** in Sampling Type. Specify the number of rows or random rows on which you want to run data profiling.
Azure Blob Container	Name of the Blob container that is used to stage profiled data. Applies to the Azure Blob storage type.
ADLS FileSystem Name	Name of the file system that is used to stage profiled data. Applies to the ADLS Gen2 storage type.

Parameter	Description
Staging File Format	File format to use when you stage the profiled data. Select one of the following formats: • **Delimited Text**. Select if the external table is created in the Delimited text file format. • **Parquet**. Select if the external table is created in the Parquet file format. Applies to external tables created on the Microsoft Azure Serverless SQL pool.
Field Delimiter	Character used to separate fields in a file. Default is `0x1e`. Non-printable characters must be specified as hexadecimal characters. **Note:** Use the field delimiter that you used while creating the external table in the Delimited text file format.
Quote Character	Specifies the quote character to skip when you read data from Microsoft Azure Synapse SQL. The quote character that you specify must not exist in the source table. If it exists, enter a different quote character value. Default is `0x1f`. **Note:** Use the quote character that you used while creating the external table in the Delimited text file format.
Maximum Precision of String Fields	The maximum precision value for profiles on string data type. You can set a maximum precision value of 255 characters. Default is 50.
Text Qualifier	The character that defines string boundaries. If you select a quote character, profiling ignores delimiters within the quotes. Select a qualifier from the list. Default is Double Quote.

6. Expand **Data Quality** and select **Enable Data Quality**.

   Note: You can click **Use Data Profiling Parameters** to use the same parameters as in the **Data Profiling** section.

   Note: Ensure that you have permissions on all the staging and flat file connections that you use in your data quality configuration. You can't run the job if you don't have permissions on the connections that you use. Select connections that you have access to, or ask the administrator to grant the necessary permissions on the connections that you want to use.

7. In the **Connection and Runtime** area, choose the Secure Agent group where you want to run catalog source jobs.

8. In the **Parameters** area, configure the parameters.

   The following table describes the properties that you can enter:

Parameter	Description
Data Quality Rule Automation	Enable the option to automatically create or update rule occurrences for data elements in the catalog source. Choose one of the following options: • **Apply on Data Elements linked with Business Dataset**. Creates rule occurrences for all data elements that are linked with business data sets in the catalog source. • **Apply on all Data Elements**. Creates rule occurrences for all data elements in the catalog source.

Parameter	Description
Data Quality Remediation	Enable the option to specify a flat file connection to store the list of failed rows so that users can remediate poor data quality scores. Choose one of the following options: • **No**. Doesn't enable the data quality failure ticket option. • **Yes**. Shows a list of flat file connections where you write failed rows to customer-managed locations.
Data Quality Failure Ticket	Specify whether you want to create data quality failure tickets for poor data quality scores based on the threshold defined for the rule occurrence in Data Governance and Catalog. Choose one of the following options: • **No**. Doesn't automatically create data quality failure tickets when the data quality scores are poor. • **Yes**. Automatically creates data quality failure tickets based on the data quality threshold values you define in Data Governance and Catalog, and notifies you when a data quality score is below the threshold. **Note:** You must configure a workflow event for the data quality failure and enable the event in Metadata Command Center.
Cache Result	Enable the option to generate a cache file in the runtime environment and preview the cached results faster in subsequent data preview runs. Choose one of the following options: • **Agent Cache**. Stores the results in the runtime environment cache for seven days by default after the first run. • **No Cache**. Doesn't store the preview results in the cache. You can only view live results.
Run Rule Occurrence Frequency	Specify whether you want to run data quality rules based on the frequency defined for the rule occurrence in Data Governance and Catalog.
Sampling Type	Determine the sample rows on which you want to run the data quality task. Choose one of the following options: • **All Rows**. Runs data quality on all rows in the metadata. • **Limit N Rows**. Runs data quality on a limited number of rows. • **Random N Rows**. Runs data quality on the selected number of random rows.
No of rows to limit	Required if you select **Limit N Rows** or **Random N Rows** in Sampling Type. Specify the number of rows or random rows on which you want to run data quality.
Azure Blob Container	Name of the Blob container that is used to stage data. Applies to the Azure Blob storage type.
ADLS FileSystem Name	Name of the file system that is used to stage data. Applies to the ADLS Gen2 storage type.

Parameter	Description
Staging File Format	File format to use when you stage the data. Select one of the following formats: • **Delimited Text**. Select if the external table is created in the Delimited text file format. • **Parquet**. Select if the external table is created in the Parquet file format. Applies to external tables created on the Microsoft Azure Serverless SQL pool.
Field Delimiter	Character used to separate fields in a file. Default is `0x1e`. Non-printable characters must be specified as hexadecimal characters. **Note:** Use the field delimiter that you used while creating the external table in the Delimited text file format.
Quote Character	Specifies the quote character to skip when you read data from Microsoft Azure Synapse SQL. The quote character that you specify must not exist in the source table. If it exists, enter a different quote character value. Default is `0x1f`. **Note:** Use the quote character that you used while creating the external table in the Delimited text file format.
Maximum Precision of String Fields	The maximum precision value for fields of the string data type. You can set a maximum precision value of 255 characters. Default is 50.
Text Qualifier	The character that defines string boundaries. If you select a quote character, data quality ignores delimiters within the quotes. Select a qualifier from the list. Default is Double Quote.

Configure data observability

Enable the data observability capability to receive anomaly notifications for metadata extracted from the source system.

To include profiling anomalies, enable data profiling for the catalog source. For accurate results, set the **Profiling Scope** option to **Full** in the **Data Profiling and Quality** configuration.

1. Click the **Data Observability** tab.
2. Select **Enable Data Observability**.
3. In the **Parameters** area, enter a value to specify the minimum number of catalog source job runs required to start detecting anomalies. For example, if you enter a value of 5, then the job starts to detect anomalies from the sixth run.

   Note: This option does not apply to anomalies that require a minimum of two runs to generate.
4. Specify the maximum number of anomaly events to generate for each catalog source run. Default is 1000.
5. Optionally, in the **Asset Filters** area, enable filters to run data observability on specific assets.
   - Specify volume filters if you want to extract volume anomaly data for specific metadata extracted.
   - Specify data profiling filters if you want to override the data profiling filter and extract profiling anomalies for assets based on a different filter.

6. To add a volume filter, perform the following steps:
   a. Select **Yes** to view filter options.
   b. From the **Include/Exclude** list, choose to include or exclude metadata.
   c. Enter a value to specify the object location.

      For example, you want to observe volume metrics of a specific table from the schema extracted. Enter the schema name followed by the table name in the input field.
      Example: `Schema_Name.Table_Name`
   d. To include or exclude multiple objects, click the **Add** icon to add filters with the OR condition.
7. To add a data profiling filter, perform the following steps:
   a. Select **Yes** to view filter options.
   b. From the **Include/Exclude** list, choose to include or exclude metadata.
   c. From the **Specify data profiling filters** list, select an object type.
   d. Enter a value to specify the object location.

      For example, you ran a profile on multiple schemas, and now you want to generate profiling anomalies on a specific schema. Enter the schema name in the input field.
   e. To include or exclude multiple objects, click the **Add** icon to add filters with the OR condition.
8. Optionally, in the **Metric Filter** area, specify filters to generate events based on specific metrics. If you don't include a filter, the job detects both profiling and other anomalies applicable to the catalog source.
   a. Select **Filter conditions**.
   b. From the **Include Metric** list, choose to include or exclude a metric.
   c. From the **All Metrics** list, select a metric for which you want to generate events.
   d. From the **Sensitivity** list, select the sensitivity of the anomaly.
   e. From the **Select detection rules** list, select rules that you apply to detect anomalies.

      The following table describes the detection rules that you can select:

Option	Description
Static Data	Detects the following anomalies: - Percentage variation. If percentage values other than 0% or 100% remain constant for three or more profile runs, any subsequent change generates a Percentage Variation anomaly. - Count variation. A change in a value that is usually constant indicates that there is an anomaly.
100% or 0% Change Detection	Detects the following anomalies: - Drop from maximum. A drop from 100% to a lower value generates a Drop from Maximum anomaly. - Surge from minimum. A spike from 0% to any higher value generates a Surge from Minimum anomaly.

Option	Description
Standard Deviation	Detects the following anomalies: - Drop in transition. A drop from a value that has been decreasing at a constant rate to a significantly lower value generates a Drop in Transition anomaly. - Surge in transition. A spike from a value that has been increasing at a constant rate to a significantly higher value generates a Surge in Transition anomaly. - Deviation. If the percentage values of a record change with each profile run, the algorithm calculates an expected range based on the changes observed in each profile run. A new value that falls outside of this expected range generates a Deviation anomaly.
Breaking Trends	Detect the following types of count-based anomalies: - Drop. A drop in a value that has been increasing to a lower value generates a Drop anomaly. - Surge. A spike in a value that has been decreasing to a higher value generates a Surge anomaly.

9. Applicable if you include data volume anomalies in the metric filters. In the **Metric Configuration** option, choose how you want to measure metadata volume.

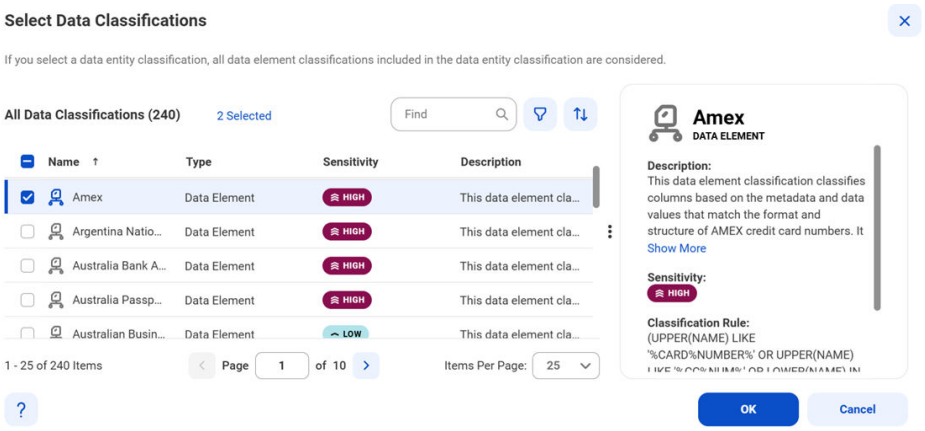
   You can choose one of the following options:

   - Statistics. Data observability volume that is extracted from the objects of the source system. Here, the volume measured might be outdated.
   - Calculated. Data observability volume that is measured when a data observability job is run on the catalog source. Here, the volume measured is more accurate.

Configure data classification

Enable the data classification capability to identify and organize data into relevant categories based on the functional meaning of the data.

1. Click the **Data Classification** tab.
2. Select **Enable Data Classification**.
3. Choose one or both of the following options:
   - **Generated Data Classifications**. CLAIRE automatically generates data classifications for the data elements.
   - **Data Classification Rules**. Choose from predefined or custom data classifications.
     1. Click **Add Data Classification**. The **Select Data Classifications** dialog box appears.



2. Select the data classifications that you want to use.
3. Click **OK**.

Configure relationship discovery

Enable the relationship discovery capability to identify pairs of similar columns and relationships between tables within a catalog source.

Before you configure relationship discovery, perform the following tasks:

- Import a relationship inference model. For more information about importing a relationship inference model, see " Import a relationship inference model" on page 17.
- Enable data profiling on the **Data Profiling and Quality** tab, and select **Keep Signatures and Values** as the run mode in the **Parameters** section. These configurations enable you to retain values of the columns in the profiling results and discover relationships.

1. Click the **Relationship Discovery** tab.
2. Select **Enable Relationship Discovery**.
3. In the **Column Similarity** area, select the **Relationship Inference Model**.

   Note: The relationship inference models that you imported appear in the **Relationship Inference Model** field.
4. In the **Joinable Tables Relationship** area, specify the **Containment Score Threshold** to identify joinable table relationships within the catalog source. This score is an indicator of the data overlap between any two given columns which determines whether the tables are joinable.

   Note: A higher score means that the objects have more overlapping data and a lower score means lesser overlapping data between the two objects. A containment score threshold lower than 0.4 might result in a large number of false positives.

After you run the catalog source job, you can view the inferred relationships on the **Relationships** tab of the extracted assets in Data Governance and Catalog.

Configure glossary associations

Enable the glossary association capability to associate glossary terms with technical assets, or to get recommendations for glossary terms that you can manually associate with technical assets in Data Governance and Catalog.

Metadata Command Center considers all published business terms in the glossary while making recommendations to associate your technical assets.

1. Click the **Glossary Association** tab.
2. Select **Enable Glossary Association**.
3. Select **Enable auto-acceptance** to automatically accept glossary association recommendations.
4. Specify the **Confidence Score Threshold for Auto-Acceptance** to set a threshold limit based on which the glossary association capability automatically accepts the recommended glossary terms.

   Note: Specify a percentage from 80 to 100. If the score is higher than the specified limit, the glossary association capability automatically assigns a matching glossary term to the data element.
5. Select **Enable Below-threshold Recommendations** to receive glossary association recommendations below the auto-acceptance threshold. If you enable auto-acceptance, you can enable below-threshold recommendations to receive glossary recommendations below the auto-acceptance threshold.

6. Specify the **Confidence Score Threshold for Recommendations** to set a threshold based on which the glossary association capability makes recommendations

   If you enable auto-acceptance, specify a percentage from 80 to the selected auto-acceptance threshold. You can accept or reject the recommended glossary terms that fall within this range in Data Governance and Catalog.

   If you disable auto-acceptance, specify a percentage from 80 to 100 inclusive.

7. Choose to automatically assign business names and descriptions to technical assets. You can then choose to retain existing assignments and only assign business names and descriptions to assets that don't have assignments, or allow overwrite of existing assignments.

   By default, existing assignments are retained.

8. Optional. Choose to ignore specific parts of data elements when making recommendations. Select **Yes** and enter prefix and suffix keyword values as needed.

   Click **Select** to enter a keyword. You can enter multiple unique prefix and suffix keywords. Keyword values are case insensitive.

9. Optional. Choose specific top-level business glossary assets to associate with technical assets. Selecting a top-level asset selects its child assets as well. Select **Top-level Glossary Assets** and specify the assets on the **Select Assets** page.

10. Optional. Choose to use abbreviations and synonym definitions from lookup tables for accurate glossary association. Select **Yes** to enable, and then click **Select** to upload a lookup table.

11. Click **Next**.

    The **Associations** page appears.

Configure automation

Enable the lineage discovery and autonomous enrichment capabilities to build complete lineage using CLAIRE and to receive enrichments propagated from catalog sources connected through lineage.

Enable Lineage-based Propagation on the Autonomous Enrichment tab of the Configure page before you configure this catalog source to receive enrichments. For more information, see *Administration*.

With the lineage-based propagation settings on this tab, you configure how this catalog source receives enrichments propagated from connected catalog sources. Enrichments propagate only to assets that don't already have enrichments.

1. Click the **Automation** tab.
2. In the **Lineage Discovery** section, select **Enable Lineage Discovery**.
3. In the **Filters** area, define one or more filter conditions to apply for lineage discovery.

   To define filters, you can choose to select catalog source types, asset groups, or enter a catalog source name or search from a list of catalog sources.

   a. Select **Yes** to view filter options.

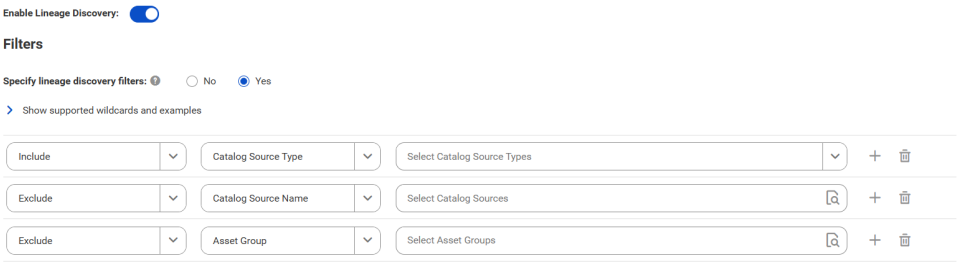
   b. From the Include/Exclude list, choose to include or exclude catalog sources for lineage discovery based on the filter parameters.

   c. From the filter type list, select catalog source type, catalog source name, or asset group.

   d. In the filter value field, select the required catalog source types, or click the Search button and select catalog sources or asset groups.

      Filters can contain the asterisk wildcard to represent multiple characters or empty text.

The filter options appear.



Examples:

- To include or exclude all Oracle catalog sources, select **Catalog Source Type** as the filter type and select `Oracle` in the filter value field.
- To include or exclude the 'Oracle_Retail' catalog source, select **Catalog Source Name** as the filter type and search for the catalog source or enter `Oracle_Retail` in the filter value field.
- To include or exclude all catalog sources with names that start with 'Oracle', select **Catalog Source Name** as the filter type and search for the catalog source or enter `Oracle*` in the filter value field.
- To include or exclude all catalog sources with names that end with 'Retail', select **Catalog Source Name** as the filter type and search for the catalog source or enter `*Retail` in the filter value field.
- To include or exclude all catalog sources with names that contain 'Ret', select **Catalog Source Name** as the filter type and search for the catalog source or enter `*Ret*` in the filter value field.
- To include or exclude all catalog sources that are part of the 'Financial Group' asset group, select **Asset Group** as the filter type and search `Financial Group` in the filter value field.

Note: You can't add more than one include or exclude filter for the same filter type.

e. Optionally, to define an additional filter with an AND condition, click the **Add** icon.

4. In the **Autonomous Enrichment** section, select **Enable Lineage-based Propagation**.

   The lineage-based propagation settings appear.

5. Select the enrichment types to receive from connected catalog sources.

   You can select Classification, Description, Business Name, and Glossary Term.

6. Choose how to apply received enrichments.
   - Select **Auto-accepted** to apply received enrichments automatically.
   - Select **Recommended** to list them as suggestions that you can curate in Data Governance and Catalog.
7. Choose whether to receive enrichments only from certified assets.

   Assets are certified in Data Governance and Catalog. This option is selected by default. If you clear it, this catalog source also receives enrichments from assets that are not certified, which might be less reliable.

8. Choose where this catalog source receives enrichments from.
   - Select **Catalog Source** to receive enrichments from a specific catalog source.
   - Select **Nearest Neighbor** to receive enrichments from the closest connected catalog source based on lineage.
9. If you select **Catalog Source**, select a catalog source from the list.

   Ensure that the catalog source you select is within 15 lineage hops. Enrichments don't propagate from a catalog source beyond that distance.

10. If you select **Nearest Neighbor**, configure the lineage settings:
    a. Set **Lineage Direction** to search upstream, downstream, or in both directions.

       Upstream catalog sources feed data into this catalog source. Downstream catalog sources receive data from this catalog source. Select both directions to search upstream and downstream at the same time.

    b. Set **Number of Hops** to define how far along lineage to search.

       The search stops at the first connected catalog source that has enrichments, up to a maximum of 15 hops.
11. Click **Save**.

When you save, the catalog source begins receiving enrichments created from that point onward. Saving doesn't retrieve enrichments created prior to this date and time. To receive such enrichments, run the catalog source job manually and include the Autonomous Enrichment capability.

Step 3. Associate stakeholders and asset groups

Associate users or user groups within a stakeholder role as stakeholders for technical assets in Data Governance and Catalog. Also, you can choose to assign technical assets extracted from the catalog source to asset groups. You can then use access policies to control permissions on assets that are assigned to asset groups.

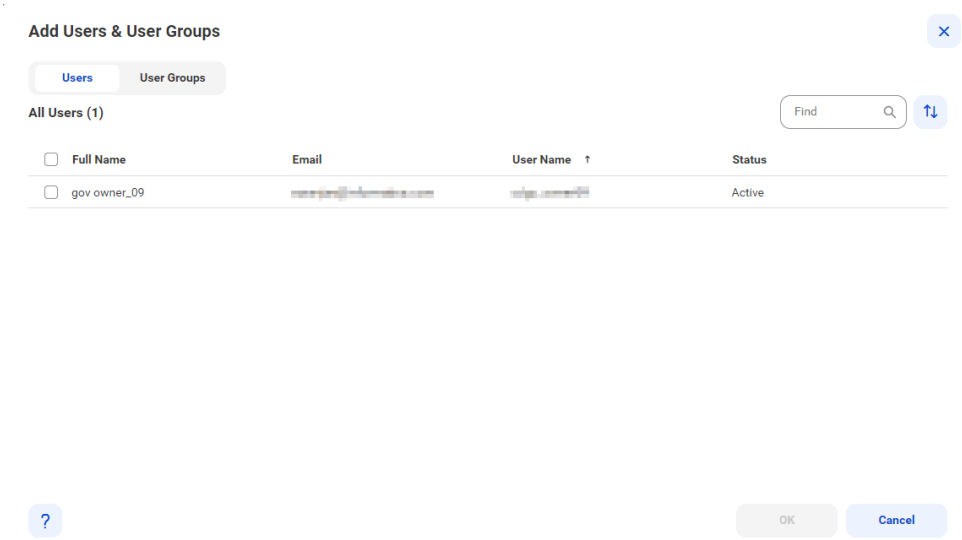
Verify that the administrator assigned users and user groups to the stakeholder role that you want to associate with technical assets.

1. To associate users or user groups as stakeholders with technical assets extracted from the catalog source, perform the following steps:
    a. On the **Associations** page, click **Stakeholders**.
    b. Select **Assign Stakeholders**.
    c. Select **Propagate to Child Assets** to propagate stakeholders of a parent asset to child assets without stakeholders.
    d. Select **Append Stakeholders** to propagate stakeholders of a parent asset to all child assets including assets with stakeholders assigned.

       Note: To enable **Append Stakeholders**, you must enable **Propagate to Child Assets**.

       You can also override the propagation settings from the catalog source page in Data Governance and Catalog. For more information about stakeholder propagation, see *Stakeholder Propagation* in *Administration*.
    e. Select a stakeholder role.
    f. Click **Select** to add users and user groups from the stakeholder role as stakeholders for the technical assets.

       The **Add Users & User Groups** dialog box displays a list of users and user groups assigned to the selected stakeholder role.



g. Select one or more users or user groups to assign as stakeholders for the technical assets, and click **OK**.

   Only the selected users and user groups belonging to the specified stakeholder role are granted the permissions to technical assets.

h. To assign users or user groups from another stakeholder role, click **Add** and then repeat the steps.

2. To assign asset groups to technical assets extracted from the catalog source, perform the following steps:

   a. On the **Associations** page, click **Asset Groups**.

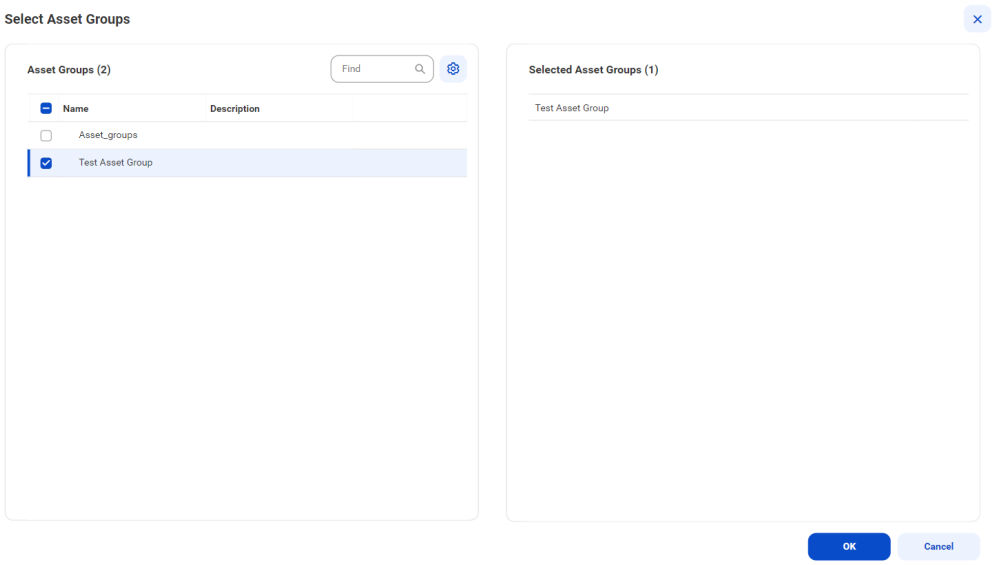
   b. Select **Assign Asset Groups**.

   c. Click **Select**.

      The **Select Asset Groups** dialog box displays the list of asset groups.

   If you enabled an access policy that includes an asset group, you can only view assets that belong to that asset group.

3. Select the asset groups to which you want to assign technical assets extracted from the catalog source, and click **OK**.



4. Choose to save and run the job or to schedule a recurring job.
   - To save and run the job, click **Save** and then **Run**.
   - To schedule a recurring job, click **Next** to open the **Schedule** page.

Step 4. Run or schedule the job

Choose to run a catalog source job manually, or configure it to run on schedule.

Note: You can't run multiple jobs simultaneously.

You can choose to perform a full or an incremental metadata extraction. A full metadata extraction extracts all objects from the source to the catalog. An incremental metadata extraction extracts only the changed and new objects since the last successful catalog source job run. Incremental metadata extraction doesn't remove deleted objects from the catalog and doesn't extract metadata of code-based objects if applicable.

When you run an incremental metadata extraction job with a filter to include metadata from objects, the job extracts only the objects that have the latest timestamp since the last successful job.

Note: If incremental metadata extraction is available for the catalog source, the option is always visible, but incremental metadata extraction jobs can run only after a scheduled full metadata extraction job. Manual jobs do not enable the incremental metadata extraction option.

Run the job manually

Click **Save** to save the catalog source and click **Run**. On the **Run Catalog Source Job** window, click **Run** to run the job.

You can override the capabilities that you selected while configuring your catalog source on the **Configuration** page. The first time you run the catalog source job, the metadata extraction capability is mandatory. From the second run onwards, you can choose to override the configured metadata change option. You can retain, delete, or deprecate objects that are deleted from the source in the catalog. For subsequent runs of the catalog source job, the metadata extraction capability is optional.

Note: You can choose incremental metadata extraction for subsequent runs only after one full metadata extraction job completes successfully. Incremental metadata extraction jobs run with the **Retain** metadata change option even if you set the option to **Delete** or **Deprecate** in the catalog source.

Note: To run a catalog source job, you need permissions on the connection to the source system. To run a catalog source job for catalog sources that reference other source systems, you need permissions on the connections for all the reference source systems.

Run the job on a schedule

You can choose to run metadata extraction and other capabilities on a recurring schedule. You can't choose incremental metadata extraction and full metadata extraction in the same schedule. To create a schedule for incremental metadata extraction, you must have completed at least one full metadata extraction job successfully. If not, first create a schedule for a full metadata extraction.

If an incremental metadata extraction is scheduled to run when the last run details aren't available, the job first performs a full metadata extraction, followed by incremental metadata extraction on subsequent runs.

For example, this can happen in the following scenarios:

- You create schedules for both incremental metadata extraction and full metadata extraction, but schedule the incremental extraction to run before the first full metadata extraction job.
- You create schedules for both incremental metadata extraction and full metadata extraction, but delete the full metadata extraction schedule before its first run.

1. On the **Schedule** tab, select **Run on Schedule**.
   The **Schedule** configuration page opens.
2. Click the checkbox corresponding to each capability that you want to include in the schedule.
3. Enter the start date, time zone, and the interval at which you want to run the job.
4. You can manage additional schedules using the following options:
   - To create a new schedule, click the **Add** button.
   - To delete a schedule, click the **Delete** button.
   - To enable or disable a schedule, click the **Enable Schedule** toggle button.

   Note: You can create a maximum of one schedule per capability that you enable. If you purged a catalog source or did not run the metadata extraction job, the catalog source job runs metadata extraction before running other scheduled capabilities.

   Note: To create a schedule, you need permissions on the connection to the source system. If you lose permissions on the connection after you create a schedule, the scheduled jobs continue to run.
5. Click **Save** to save the schedule.

Monitor job status

After the job runs, you can monitor the status of the job on the **Overview** page of the job.

For more information about job monitoring, see *Administration*.

Step 5. Assign reference catalog source connections to endpoint catalog source objects

When you run the catalog source job, if the catalog source references another source system, a reference catalog source and connection get created that point to the reference source system. To view the complete

lineage for your catalog source, you can perform connection assignment from the reference catalog source connection to the objects in the reference source system. A reference source system might be a database, such as Microsoft Azure Synapse Data Warehouse. You must first create and run an endpoint catalog source that connects to the reference source system.

Before you assign a connection, ensure that you have created and run an endpoint catalog source for each reference source system.
Note: If the source schema contains case-sensitive tables or if the reference objects contain multiple objects with the same name in different cases, perform case-sensitive connection assignment to get correct lineage.

If you enabled the lineage discovery capability for your catalog source, you can either curate the CLAIRE recommended endpoint objects on the **Related Catalog Sources** tab or assign connections manually.

For more information about related catalog sources and lineage discovery, see *Lineage discovery* in the *Administration* help.

1. On the **Configure** page, select the **Lineage** tab, and then select the **Lineage Discovery** tab. On the **Catalog Sources** panel, select the required catalog source and click the **Assign Connections** tab.

   The **Assign Connections** tab displays a list of assigned and unassigned connections along with details for each connection. Use filters to view the connections based on the connection names. Click the **Add Filter** menu to add filters.

2. Select the connection to the reference source system and click **Assign**.

   The connection name appears prefixed to the reference catalog source name on the **Hierarchy** tab of your catalog source in Data Governance and Catalog.

   The **Assign Connection** dialog box appears with a list of recommended objects from the endpoint catalog sources. Click **All** to view all endpoint catalog source objects.

3. In the **Assign Connection** dialog box, select one or more objects from the endpoint catalog sources and click **Assign**.

   You can filter the list in the **Assign Connection** dialog box by name, type, or endpoint.

   You can assign a Microsoft Azure Synapse Data Warehouse or Microsoft Azure Data Factory source system as a referenced source system. The objects must belong to the Database class type.

When you click **Assign**, Metadata Command Center creates links between matching objects in the connected catalog sources, and it calculates the percentage of matched and unmatched objects. The higher the percentage of matched objects, the more accurate the lineage that you view in Data Governance and Catalog.

CHAPTER 4

View results in Data Governance and Catalog

After Metadata Command Center runs a job, you can view the results in Data Governance and Catalog where the catalog source and its elements are called technical assets. You can view a catalog source as a hierarchy. Expand each technical asset to see its components.

When referenced source systems are connected to a catalog source, you can expand the hierarchy to see details about the technical asset's component elements.

You can view the data lineage of an asset contained within a catalog source to see individual elements such as data sources, calculations, and filters. When you view data lineage, you can see the individual upstream elements that contribute data or expressions to each component of a data flow or catalog source.

View metadata extraction results

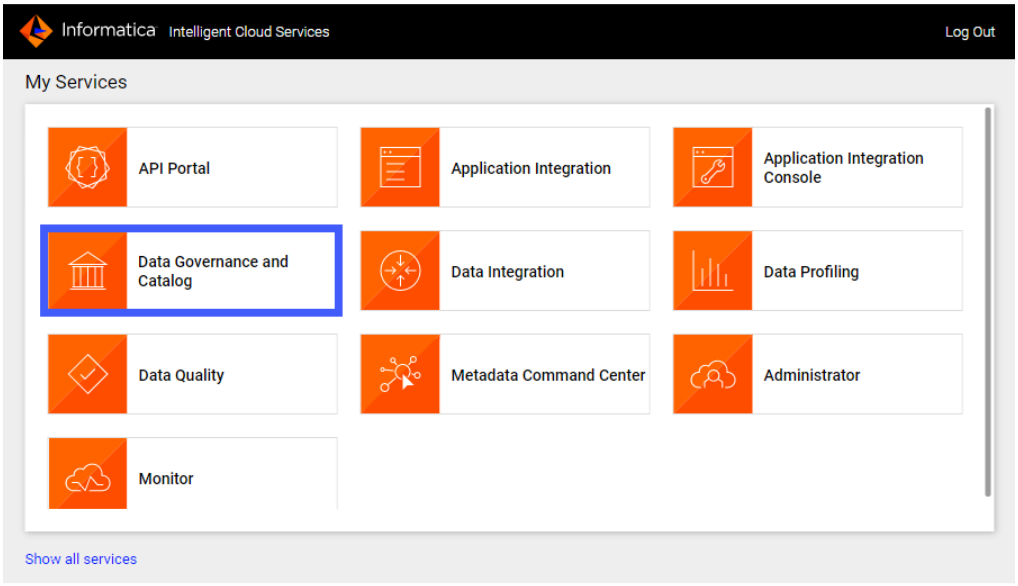
After a job runs in Metadata Command Center, view the results in Data Governance and Catalog. You can view details about source system contents in a hierarchical structure and trace data lineage.

1. Log in to Intelligent Data Management Cloud.

   The **My Services** page appears.

2. Click Data Governance and Catalog.

   The Data Governance and Catalog box appears on the **My Services** page.



3. On the Data Governance and Catalog home page, click the number in the **Technical Assets** panel.

   The **Technical Assets** page opens.

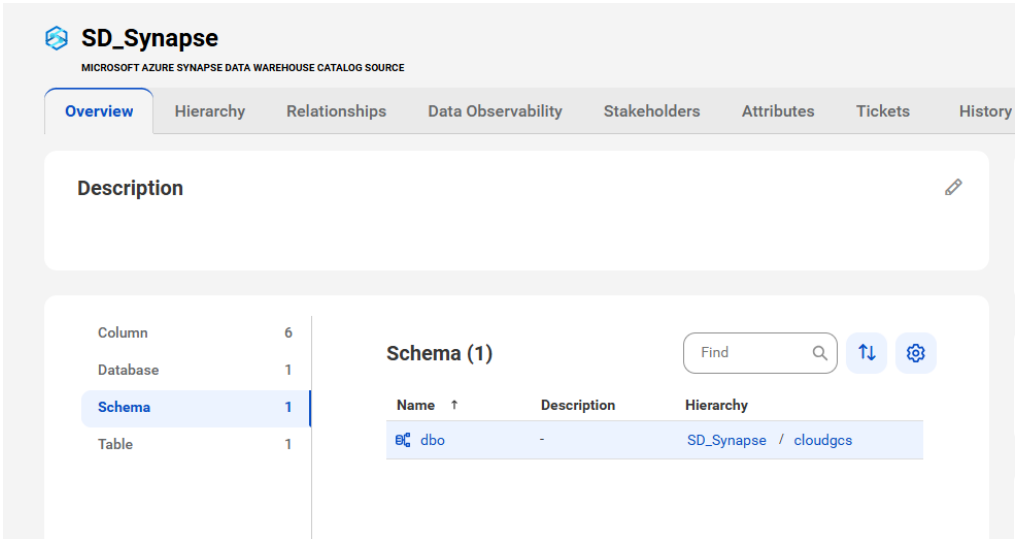
4. Select **Catalog Source** in the **Filter** list.

   The list of catalog sources opens.

5. Search for the catalog source from which you extracted metadata, and click the name.

   The **Overview** tab of the asset opens.

   The following image shows a sample Microsoft Azure Synapse Data Warehouse catalog source asset page:



6. View the asset from different perspectives by clicking on the tabs.

   Note: If a new record is added to the source system and you run an incremental metadata extraction job, no objects are extracted because the metadata has not changed. If an incremental metadata extraction job includes a view, the job extracts the corresponding tables even if the tables were extracted in previous runs.

For more information about working with assets, see Working with Assets in *Data Governance and Catalog* help.

View data lineage

Data lineage is a visual representation of the flow of data across the systems in your organization. Lineage depicts how the data flows from the system of its origin to the system of its destination.

Data lineage views are available for technical assets in the catalog source. You can view lineage at the catalog source, data set, or data element level.

The lineage at the catalog source level shows how data flows from one catalog source to another. The lineage at the data set and the data element levels show how other technical assets such as files or tables contribute to the selected asset.

If linking catalog sources is available for your catalog source, you can use Metadata Command Center to generate data lineage based on rules or by generating automated lineage with CLAIRE. You can choose source and target catalog sources and objects to link and generate lineage.

To determine whether linking catalog sources is available for your catalog source, navigate to the **Configuration** tab of the **Link Catalog Sources** page. The catalog source must appear in the list of source and target catalog sources.

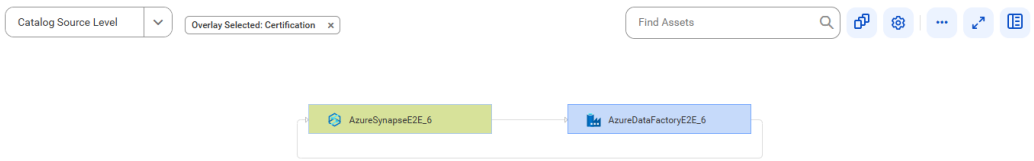
For information about linking catalog sources, see *Link catalog sources* in the Administration help.

View lineage at the catalog source level

The catalog source level shows how data flows from one catalog source to another with the lineage aggregating data from the data set and data element levels.

To view data lineage at the catalog source level, open a technical asset, click the **Lineage** tab, and then verify that the level is set to **Catalog Source Level**.

The following image shows how the AzureDataFactoryE2E_6 catalog source gets data from the AzureSynapseE2E_6 catalog source after connection assignment:



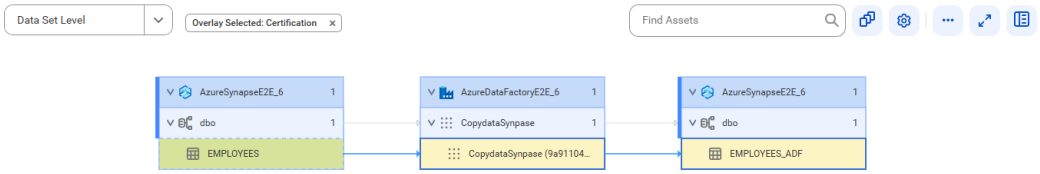
After connection assignment, the referenced object icons change to specific object icons.

View lineage at the data set level

The data set level displays individual sets of data in the data flow.

To view lineage at the data set level, open a technical asset, click the **Lineage** tab, and then verify that the level is set to **Data Set Level**.

The following image shows data set level lineage where the EMPLOYEES_ADF target table gets data from the EMPLOYEES source table using the CopydataSynapse (9a91104a) activity instance after connection assignment:



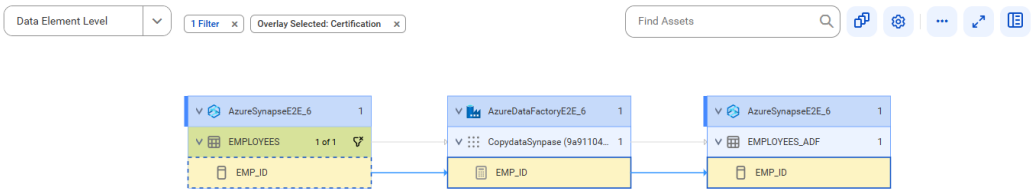
After connection assignment, the referenced object icons change to specific object icons.

View lineage at the data element level

The data element level displays details of the data set level. At the data element level, you can see the input sources for expressions or commands and calculations or transformations on the data.

To view data lineage at the data element level, open a technical asset, click the **Lineage** tab, and then verify that the level is set to **Data Element Level**.

The following image shows data element level lineage where the EMP_ID column of the EMPLOYEES_ADF table gets data from the EMP_ID column of the EMPLOYEES table using the EMP_ID calculation of the CopydataSynapse (9a91104a) activity instance after connection assignment:



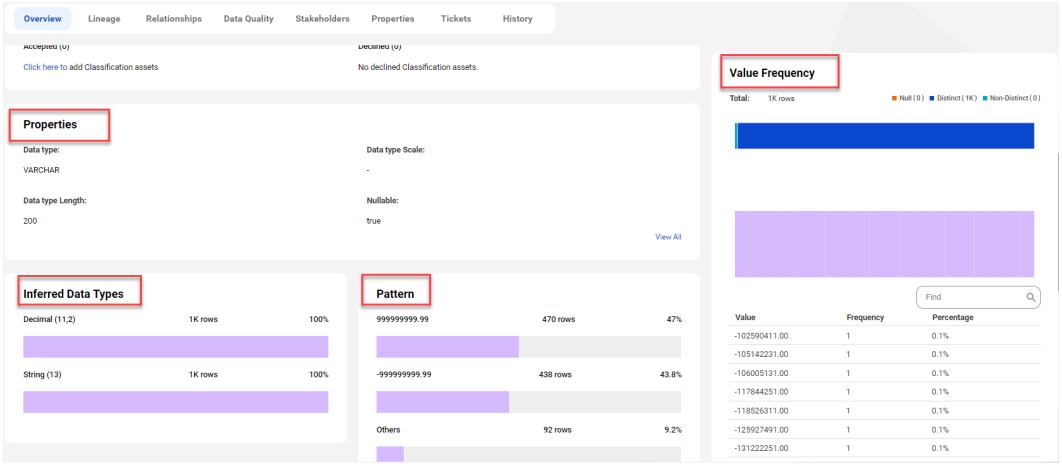
After connection assignment, the referenced object icons change to specific object icons.

View data profiling results

When you enable the data profiling task for a catalog source in Metadata Command Center, the system runs a profile to evaluate the quality of the metadata extracted from the source system. The profiling statistics appear in Data Governance and Catalog when you open the technical assets.

The scope of profiling statistics that Data Governance and Catalog displays depends on the data profiling configuration parameters that you set when you configured the catalog source in Metadata Command Center.

The following image shows the data profiling statistics that appear on a column asset page in Data Governance and Catalog:



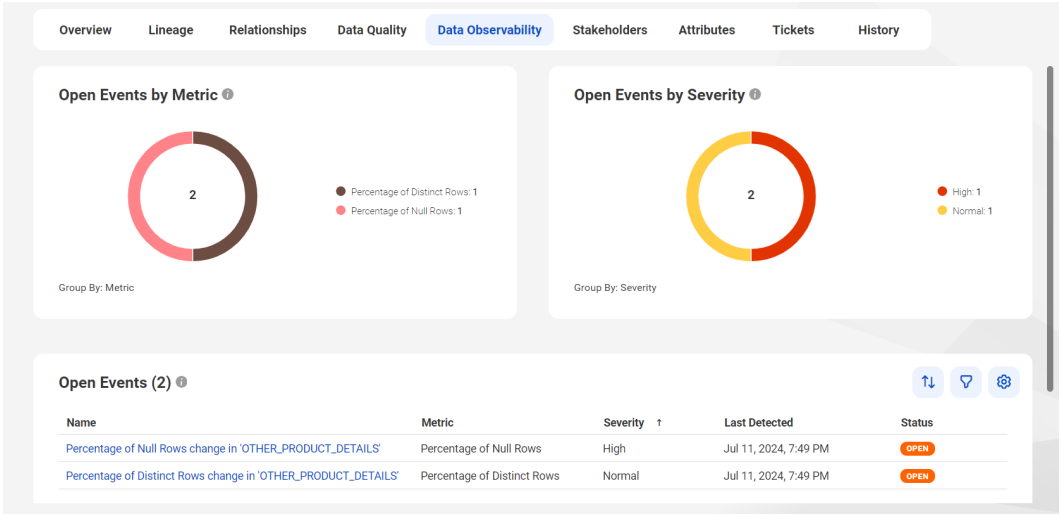
For more information about data profiling results, see *Asset Details* in the Data Governance and Catalog help.

View data observability results

When you enable data observability for a catalog source in Metadata Command Center, you can view and evaluate the events that it generates in Data Governance and Catalog. These events indicate anomalies identified in the characteristics of the profiled data in your source system.

You can view the events that data observability generates for anomalies identified for catalog sources, technical data sets, and data elements. You can then take appropriate actions for the generated events.
Note: The administrator of the catalog source might have applied filters to the data to narrow down the data elements that are applicable for business users in Data Governance and Catalog. The data for which users receive anomaly notifications depend on the filters that are configured for the catalog source.

The following image shows the open events for a column asset in Data Governance and Catalog:

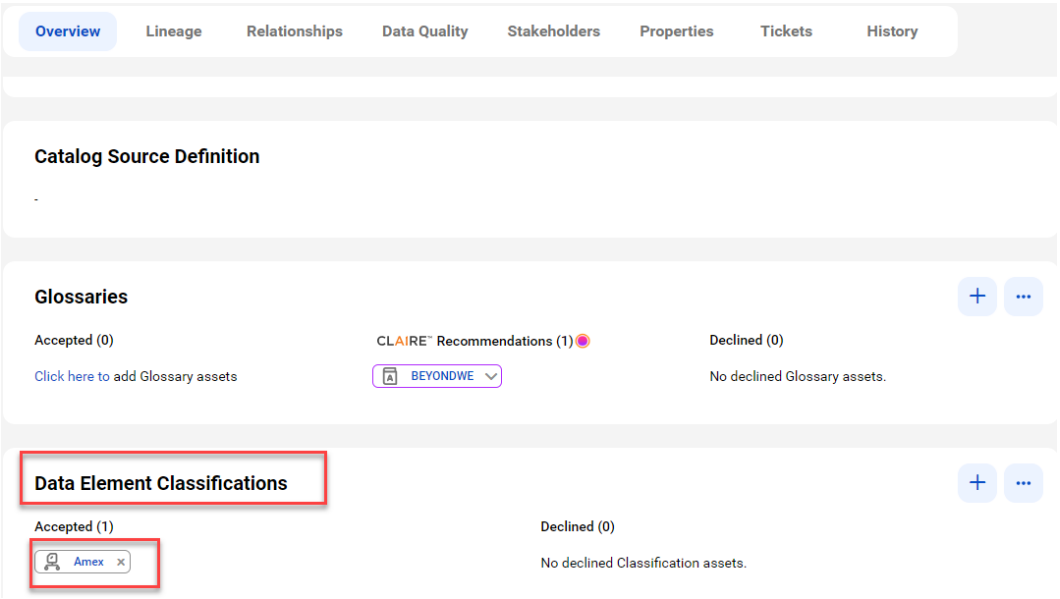


For more information about data observability results, see *Working With Assets* in the Data Governance and Catalog help.

View classified data

When you add data classification rules to a catalog source in Metadata Command Center, the system identifies the columns and tables that match the rules and displays one or more matched data classifications on the column or table asset pages in Data Governance and Catalog.

The following image shows a column asset page with the inferred data element classifications that match the column data and metadata:



For more information about data classification assets, see *Asset Details* in the Data Governance and Catalog help.

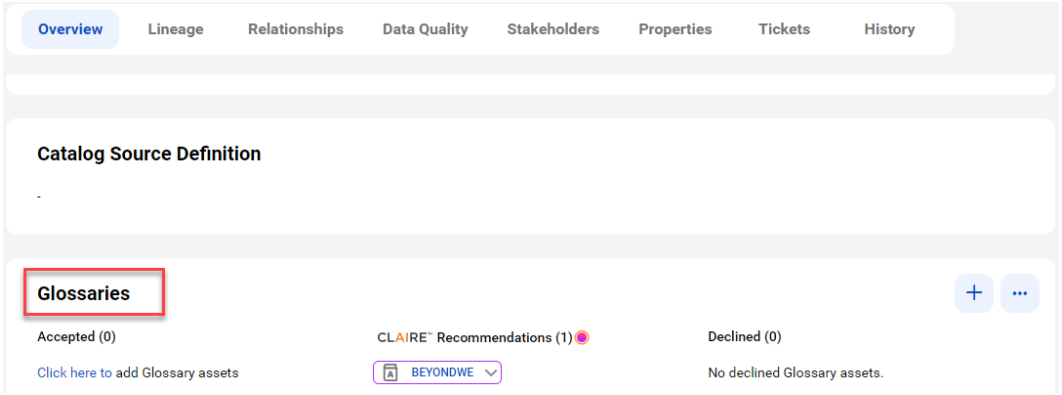
View glossary associations

When you enable the glossary association capability for a catalog source in Metadata Command Center, you can view the accepted glossary assets in Data Governance and Catalog.

The **Overview** tab for a technical asset in the catalog source displays glossary assets in the Accepted and CLAIRE Recommendations sections.

The **Glossaries** panel shows the automatically accepted and CLAIRE® recommended terms.

The following image shows a sample asset page:



APPENDIX A

Managing catalog sources

You can update, copy, delete, purge, and run a catalog source. You can also grant configuration permissions for catalog sources to different roles in your organization.

On the **Catalog Sources** page of the **Explore** page, you can select a catalog source to see its overview and related catalog sources. The **Overview** tab displays the capabilities, filters, stakeholders, asset groups, and schedules enabled for a catalog source. The **Related Catalog Sources** tab displays a list of related catalog sources that are either assigned manually using connection assignment or are curated based on CLAIRE recommendations using the lineage discovery capability. You can view related catalog sources and accept or reject CLAIRE recommendations for catalog sources that are run with the lineage discovery capability.

Cloning a catalog source configuration

You can clone the configuration of a catalog source to create another catalog source with the same configuration.

Note: Cloning doesn't copy the content of the catalog source.

1. In Metadata Command Center, go to the **Explore** page.
2. Select **Catalog Sources** from the menu.
3. You can clone a catalog source in one of the following ways:
   - Hover your mouse over the catalog source that you want to clone and click **Clone** from the **Action** menu.
   - Open the catalog source that you want to clone and click **Clone**.

   Note: By default, the cloned catalog source has the same name suffixed with _Clone. You can change the name if needed.
4. If needed, modify the configuration of the catalog source by updating the parameters on each tab.

   Cloning doesn't copy the content of sensitive fields. If the catalog source contains sensitive fields such as Client Secret, enter the values.
5. Click **Save**.

Deleting or purging a catalog source

Delete or purge a catalog source based on your business requirements.

When you delete a catalog source, you delete all the configuration specific to that catalog source, along with all the metadata that was extracted and any enrichment made to the associated metadata. Purging a catalog source deletes all the metadata that was extracted, along with any enrichment made to the associated metadata.

Note: Before you delete or purge a catalog source, manually unassign the connections that are assigned to that catalog source.

1. Go to the **Explore** page in Metadata Command Center.
2. Select **Catalog Sources** from the menu.
3. From the list of catalog sources, search for the catalog source that you want to delete or purge.
4. Click the **Action** menu, and select **Delete** or **Purge**.

   Note: You can't delete or purge a catalog source if the catalog source job is in the running state.
5. To confirm delete or purge, click **OK**.

   The catalog source is triggered to delete or purge. You can monitor the status of the job on the **Monitor** page.

   Note: If the deletion job succeeds, the catalog source disappears from Metadata Command Center. If it fails, some associated objects might still be present in Data Governance and Catalog. Such catalog sources continue to appear in the list of catalog sources in Metadata Command Center, but are marked as **Deleted**. Delete the catalog sources to remove them along with any remaining objects present in Data Governance and Catalog.

Purging obsolete objects from a catalog source

You can purge obsolete objects from a catalog source.

When you configure or run a catalog source, you can choose to deprecate objects that are deleted from a source system. If you delete objects from the source system or make changes to the filter, the objects imported into the catalog before the change move to the Obsolete lifecycle in the catalog. You can purge obsolete objects from a catalog source. This permanently deletes all objects with the Obsolete lifecycle status, along with associated enrichments, from the catalog.

1. Go to the **Explore** page in Metadata Command Center.
2. Select **Catalog Sources** from the menu.
3. From the list of catalog sources, search for the catalog source from which you want to purge obsolete objects.
4. Click the **Action** menu, and select **Purge Obsolete Objects**.

   A warning message appears on the page.
5. Click **Yes** to confirm.

   A confirmation message appears on the page and the purge job starts. You can monitor the status of the job on the **Monitor** page. If the job succeeds, all obsolete objects present in the catalog source disappear from the catalog.

   Note: If the job fails, some obsolete objects might remain in the catalog. To remove the remaining obsolete objects, run the purge job again.

Curating CLAIRE recommendations

You can view related catalog sources and accept or reject CLAIRE recommendations for a catalog source.

Applicable to catalog sources that generate lineage.

1. Go to the **Explore** page in Metadata Command Center.
2. Select **Catalog Sources** from the menu.
3. From the list of catalog sources, search for the catalog source for which you want to curate CLAIRE recommendations.
4. Hover your mouse over the catalog source and click **View Lineage Discovery** from the **Action** menu.

   Note: The **View Lineage Discovery** option is available for catalog sources that are run with the lineage discovery capability.

   The **Configure** page appears with the catalog source filtered on the **Lineage Discovery** tab of the **Lineage** tab.
   The **Related Catalog Sources** tab displays a list of related catalog sources.
5. Hover your mouse over a related catalog source and click **Accept** or **Reject** from the **Action** menu.

   For more information about lineage discovery and curating CLAIRE recommendations, see *Lineage discovery* in the *Administration* help.

Managing configuration permissions for catalog sources

As a service administrator, you can grant configuration permissions for catalog sources to different roles in your organization, and add specific users and user groups to those roles so that they can perform operational activities on a catalog source.

To manage permissions for catalog source configuration, verify that at least one role has the catalog source configuration **Set Permissions** functionality enabled for Metadata Command Center in Administrator. In Administrator, create the required roles, assign users and user groups to the roles, and specify appropriate permissions to the roles.

You can select roles and add specific users or user groups to those roles to grant the following levels of permission for catalog source configuration:

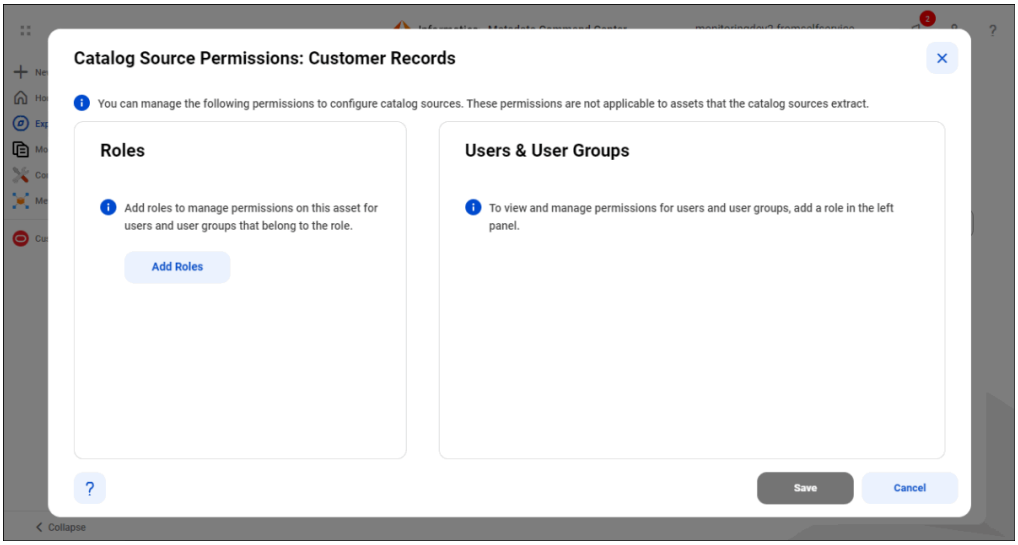
- Create
- Read
- Update
- Run
- Set Permissions

Based on the permissions granted to the roles, the selected users and user groups belonging to that role can perform appropriate operations for configuring catalog sources.

1. Go to the **Explore** page in Metadata Command Center.
2. From the list of catalog sources, search for the catalog source for which you want to manage permissions.

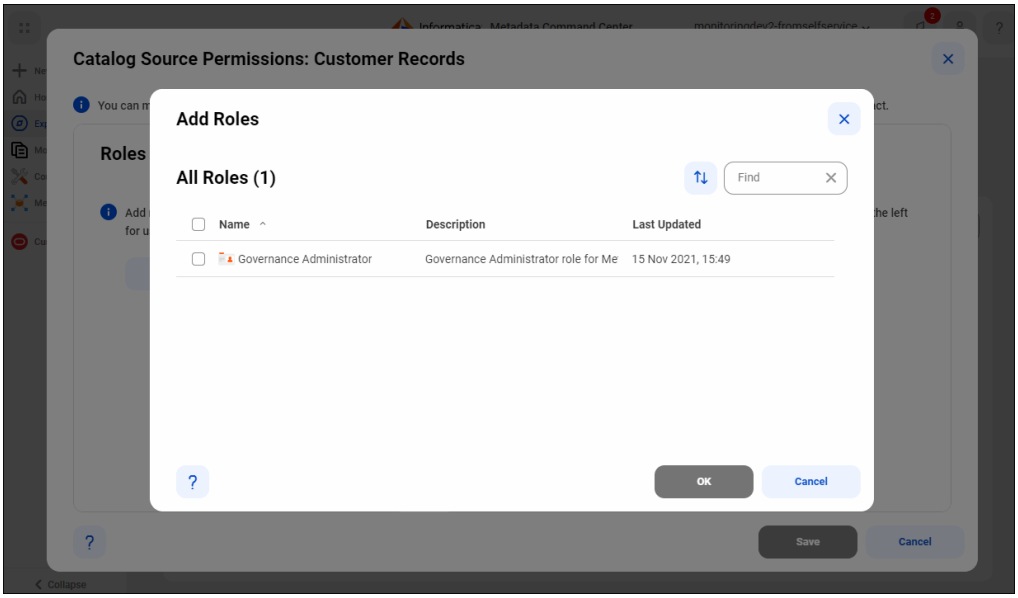
3. Click the **Action** menu, and select **Permissions**.

   The **Catalog Source Permissions** dialog box appears.



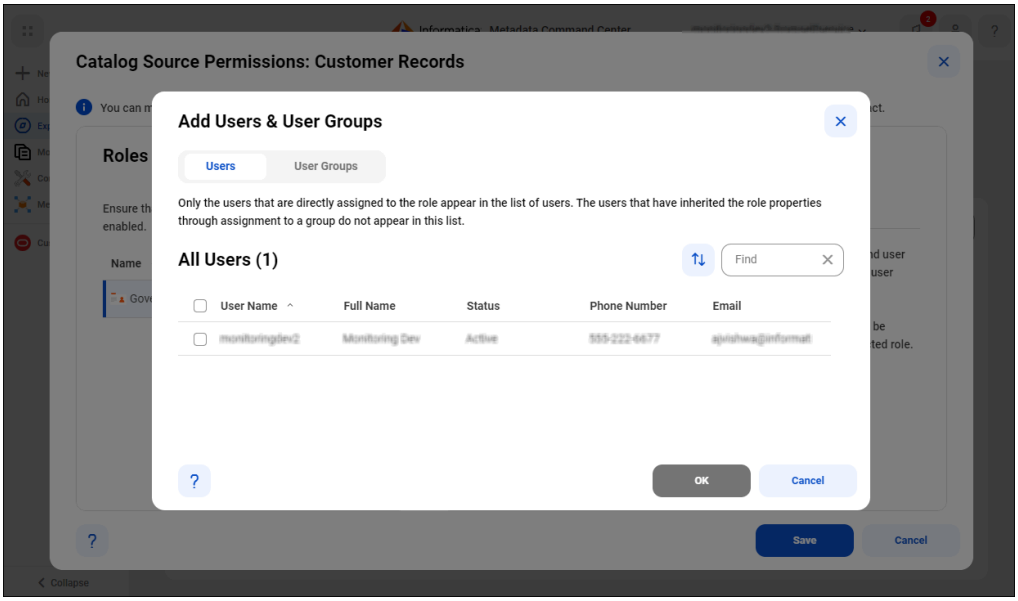
4. In the **Roles** panel, click **Add Roles**.

   The **Add Roles** dialog box appears.



5. Select one or more roles, and click **OK** to add the roles.
6. In the **Add Users & User Groups** panel, by default, all users and user groups that belong to the selected role are granted the role-defined permissions for the catalog source configuration.

To assign permissions only to the selected users and user groups that belong to this role, choose **Specific Users and User Groups**.



7. Select **Add Specific Users and User Groups**.

   The **Add Users & User Groups** dialog box displays the list of users and user groups that the administrator has assigned to the selected role.

8. Select one or more users or user groups to assign to the selected role, and click **OK**.

   Only the selected users and user groups belonging to the specified role will be granted the role-defined permissions for catalog source configuration.

9. View the assigned permissions for each user and or group, and click **Save**.